21EIC552J
INDUSTRIAL PROCESS AUTOMATION
ASSIGNMENT – I

PLC PROGRAMMING AND APPLICATIONS

Name	:	Bharathwaj Manoharan,
Register No.	:	RA2512053010003,
Dept	:	Automation and Robotics,
Year/Semester	:	First Year/First Semester.

1. **AIM:**

To design and develop ladder logic programs for a wide range of industrial automation problems, from basic combinational logic to complex sequential processes involving timers, counters, and arithmetic operations.

2. **PILOT LIGHT COMBINATIONAL LOGIC:**

2.1. PROBLEM STATEMENT:

Design a ladder logic program to energize a pilot light if, and only if, all of the following conditions are met simultaneously:

a) All four pressure switches are closed.
b) At least two out of three limit switches are closed.
c) The (Normally Closed) reset switch is not being pressed.

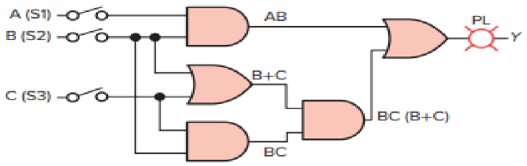
2.2. LADDER DIAGRAM:

2.3. I/O MAPPING:

3. LOGIC CIRCUIT ANALYSIS (A):

3.1. PROBLEM STATEMENT:

For the given logic gate circuit (Below Figure), determine the governing Boolean equation for output Y and subsequently develop the equivalent ladder logic program.



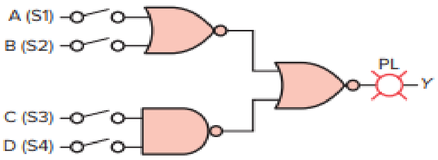
3.2. LADDER DIAGRAM:

3.3. I/O MAPPING:

4. LOGIC CIRCUIT ANALYSIS (B):

4.1. PROBLEM STATEMENT:

For the given logic gate circuit (Figure 2, containing NOR and NAND gates), determine the governing Boolean equation for output Y and subsequently develop the equivalent ladder logic program.



4.2. LADDER DIAGRAM:

4.3. I/O MAPPING:

5. FAN CONTROL LOGIC:

5.1. PROBLEM STATEMENT:

Develop a ladder logic program to control a fan based on a precise set of logical conditions. The fan must energize only when all of the following are true:

a) Input A is OFF.
b) (Input B is ON) OR (Input C is ON).
c) (Input D is ON) AND (Input E is ON).
d) (Input F is ON) OR (Input G is ON) OR (Input H is ON).

5.2. LADDER DIAGRAM:

5.3. I/O MAPPING:

6. DRILL PRESS SAFETY INTERLOCK:

6.1. PROBLEM STATEMENT:

Develop a ladder logic program for a "two-hand" safety interlock on a drill press. The drill motor must energize only when three inputs are simultaneously active: Switch 1 (PB1), Switch 2 (PB2), and the Part Sensor.

6.2. LADDER DIAGRAM:

6.3. I/O MAPPING:

7. CONTINUOUS FILLING OPERATION:

7.1. PROBLEM STATEMENT:

Design a ladder logic program for a continuous filling operation. The system must:

a) Use Start/Stop buttons to latch the conveyor motor and control 'Run'/'Standby' status lights.
b) Stop the conveyor when a photo sensor detects a box.
c) With the conveyor stopped, open a solenoid valve to fill the box.
d) Close the valve when a level sensor is activated.
e) Energize a 'Full' light, which must remain ON until the box moves clear of the photo sensor (Low/High).

7.2. LADDER DIAGRAM:

7.3. I/O MAPPING:

8. DUAL TANK FILLING SYSTEM:

8.1. PROBLEM STATEMENT:

Design a ladder logic program for an automatic dual tank filling system. A manual Start button shall run a pump to fill Tank 1 via Solenoid 1. When Sensor 1 (Tank 1 Full) activates, Solenoid 1 must close, Solenoid 2 must open (to fill Tank 2), and an indicator lamp for Tank 1 must light. When Sensor 2 (Tank 2 Full) activates, the pump must stop, and an indicator lamp for Tank 2 must light.

8.2. LADDER DIAGRAM:

8.3. I/O MAPPING:

9. THREE-STATION MOTOR CONTROL:

9.1. PROBLEM STATEMENT:

Develop a ladder logic program to control a single motor from three separate locations. Each location provides a Normally Open (NO) 'Start' pushbutton and a Normally Closed (NC) 'Stop' pushbutton. The motor must be able to be started or stopped from any of the three stations.

9.2. LADDER DIAGRAM:

9.3. I/O MAPPING:

10. On-Delay Timer Application:

10.1. PROBLEM STATEMENT:

Implement a ladder logic program using an On-Delay Timer (TON). An output light (PL) must energize exactly 15 seconds after a maintained input switch (S1) is turned ON. The light must de-energize immediately if S1 is turned OFF.

10.2. LADDER DIAGRAM:

10.3. I/O MAPPING:

11. AUTOMATED LIQUID MIXING PROCESS:

11.1. PROBLEM STATEMENT:

Design a ladder logic program for an automated batch mixing process. A Start button shall initiate the sequence:

a) Solenoid A (Fill) energizes until a 'Full' sensor is triggered.
b) Solenoid A de-energizes, and an 'Agitate Motor' runs for 3 minutes.
c) The motor stops, and Solenoid B (Drain) energizes until an 'Empty' sensor is no longer triggered.
d) The system must then wait for the next Start command.

11.2. LADDER DIAGRAM:

11.3. I/O MAPPING:

12. DELAYED LIGHTING CONTROL:

12.1. PROBLEM STATEMENT:

Develop a ladder logic program for a cascading delayed lighting sequence. When a 'Main Lights' switch is turned OFF, an 'Exit Door Light' must remain ON for an additional 2 minutes. After the 2-minute timer for the exit light elapses, a 'Parking Lot Light' must remain ON for an additional 3 minutes before turning off.

12.2. LADDER DIAGRAM:

12.3. I/O MAPPING:

13. SEQUENTIAL TAILLIGHT SYSTEM:

13.1. PROBLEM STATEMENT:

Develop a ladder logic program to control a 3-light sequential taillight system (L1, L2, L3) using a single input switch. The sequence must be: L1 ON (1s), then L1+L2 ON (1s), then L1+L2+L3 ON (1s), then all OFF (1s). This 4-second cycle must repeat as long as the input switch is active. The system must not operate if both Left and Right turn switches are active.

13.2. LADDER DIAGRAM:

13.3. I/O MAPPING:

14. CASCADING CLOCK COUNTER:

14.1. PROBLEM STATEMENT:

Construct a ladder logic program to create a cascading counter. A timer must increment a 'Minute’s' counter (C1) by one count every 60 seconds. When C1 reaches a preset of 60, it must reset and simultaneously increment an 'Hour’s' counter (C2). When C2 reaches a preset of 12, it must also reset.

14.2. LADDER DIAGRAM:

14.3. I/O MAPPING:

15. FLASHER WITH FINITE COUNT:

15.1. PROBLEM STATEMENT:

Design a ladder logic program initiated by a momentary push button. The program must flash a light (2 seconds ON, 2 seconds OFF) and increment a counter after each full flash cycle. The entire sequence must stop automatically after 4 flashes (counts) are completed, and the counter must reset.

15.2. LADDER DIAGRAM:

15.3. I/O MAPPING:

16. CONVEYOR PART TOTALIZER:

16.1. PROBLEM STATEMENT:

Construct a ladder logic program that uses math instructions to calculate a total part count. The program must, at 15-second intervals (controlled by a timer), read the integer values from three separate input registers (Count 1, Count 2, Count 3), compute their sum, and store the result in a 'Total Count' register.

16.2. LADDER DIAGRAM:

16.3. I/O MAPPING:

17. COMPARATOR LOGIC (OUT-OF-RANGE):

17.1. PROBLEM STATEMENT:

Develop a ladder logic program using comparator instructions. An output must be energized if, and only if, a value in a data register (e.g., a counter's accumulator) is outside a specific range: less than 34 OR greater than 41.

17.2. LADDER DIAGRAM:

17.3. I/O MAPPING:

18. COUNTER-CONTROLLED LATCH (IN-RANGE):

18.1. PROBLEM STATEMENT:

Design a ladder logic program to control an output based on a counter's value. The output light must turn ON (latch) when a counter's value reaches 23 and turns OFF (unlatch) when the same counter's value reaches 31.

18.2. LADDER DIAGRAM:

18.3. I/O MAPPING:

19. COUNTER-TIMER COMBINATION:

19.1. PROBLEM STATEMENT:

Construct a ladder logic program combining a counter and a timer. When a counter monitoring parts reaches a preset of 25, it must trigger an On-Delay Timer. This timer, in turn, shall run a 'Paint Sprayer' output for exactly 40 seconds.

19.2. LADDER DIAGRAM:

19.3. I/O MAPPING:

20. BATCH COUNTING AND DIVERTER:

20.1. PROBLEM STATEMENT:

Design a ladder logic program for a batch-counting and diverter system. A proximity sensor shall increment a counter for each part on a conveyor. When the counter's accumulated value reaches 1000, the program must activate a 'Gate Solenoid' for 2 seconds (controlled by a timer) to divert that part. The counter must then automatically reset and resume counting.

20.2. LADDER DIAGRAM:

20.3. I/O MAPPING:

21. SMART ROOM LOGIC TABLE:

21.1. PROBLEM STATEMENT:

Develop a ladder logic program using comparator instructions to implement a 'Smart Room' control table. The logic must read an integer value ('No. of Persons') from a register and control 'Fans' and 'Lights' outputs according to two conditions:

a) If the value is between 1 and 5 (inclusive), 1 Fan and 1 Light are ON.
b) If the value is between 6 and 10 (inclusive), 3 Fans and 3 Lights are ON.

21.2. LADDER DIAGRAM:

21.3. I/O MAPPING:

22. RESULT:

Thus, ladder logic programs for various applications using timers, counters, arithmetic operations, and sequential logic were successfully designed and prepared for testing.